Computer Science practice paper: answers and marks

How to mark your paper. Go through one question at a time. Read the answer, then check your own answer against "How the marks are given". Give yourself the mark if your answer means the same thing, even if you used different words.

Write your mark next to each question, then add them up. Read the "Watch out" boxes even when you got the marks: they're the mistakes that cost people marks in the real exam.

1(a)

1 mark

State the purpose of the CPU.

The answer

The CPU fetches, decodes and executes instructions.

How the marks are given

- 1 mark: fetch, decode and execute instructions, or process data and instructions.

Watch out

- "It stores data" is wrong: that's memory's job. "The brain of the computer" isn't enough on its own.

1(b)

1 mark

Which register holds the address of the next instruction to be fetched? Tick one box.

The answer

The Program Counter (PC).

How the marks are given

- 1 mark: Program Counter.

Watch out

- The MAR holds the address being used right now, not the next one.

1(c)

2 marks

Explain how increasing the number of cores in a CPU can improve its performance.

The answer

Each core can fetch, decode and execute instructions on its own, so more instructions can be processed at the same time.

How the marks are given

- 1 mark: you say each core can process instructions by itself.
- 1 mark: you say this means more instructions or tasks can be done at the same time.

Watch out

- It only helps if the program can be split up to run on more than one core.

1(d)

1 mark

Give one example of an embedded system.

The answer

A washing machine (or a microwave, dishwasher, central heating controller, car engine management system, traffic lights).

How the marks are given

- 1 mark: any device with a computer built in to do one job.

Watch out

- A laptop, phone or tablet isn't an embedded system. They're general-purpose computers.

2(a)

2 marks

Explain two differences between RAM and ROM.

The answer

RAM is volatile, so it loses its contents when the power goes off, but ROM is non-volatile and keeps its contents. RAM can be read from and written to, but ROM can only be read.

How the marks are given

- 1 mark for each difference, up to 2 marks. Each one must say what RAM is like and what ROM is like.
- Differences that count: volatile or non-volatile; read and write or read-only; holds the programs in use or holds the start-up instructions; RAM is bigger.

Watch out

- Saying only what RAM is like isn't a difference. Compare both.

2(b)

1 mark

Convert the binary number 10110110 into denary.

The answer

182. Working: 128 + 32 + 16 + 4 + 2 = 182.

How the marks are given

- 1 mark: 182.

Watch out

- Write the place values over the bits first: 128, 64, 32, 16, 8, 4, 2, 1.

2(c)

1 mark

Convert the denary number 75 into an 8-bit binary number.

The answer

01001011. Working: 75 = 64 + 8 + 2 + 1.

How the marks are given

- 1 mark: 01001011.

Watch out

- The question asks for 8 bits, so you need the 0 at the front. 1001011 on its own gets no mark.

2(d)

1 mark

Convert the hexadecimal number 3F into denary.

The answer

63. Working: 3 × 16 = 48, and F = 15, so 48 + 15 = 63.

How the marks are given

- 1 mark: 63.

Watch out

- Remember the hex letters: A is 10, B is 11, C is 12, D is 13, E is 14, F is 15.

2(e)

2 marks

Add the binary numbers 01101100 and 00110101. Give your answer as an 8-bit binary number. Show your working.

The answer

10100001. Working: add the columns from the right, carrying 1 when 1 + 1 = 10. Check: 01101100 is 108 and 00110101 is 53, and 108 + 53 = 161, which is 10100001.

How the marks are given

- 1 mark: the working, with the carries shown (or a denary check that adds up).
- 1 mark: the answer, 10100001.
- The question asks you to show your working, so the answer on its own only gets 1 mark.

Watch out

- Line the bits up from the right, and carry the 1 when 1 + 1 = 10. Check your answer by converting to denary.

2(f)

2 marks

An image is 100 pixels wide and 50 pixels high. It has a colour depth of 8 bits. Calculate the file size of the image in bytes. Ignore metadata. Show your working.

The answer

5000 bytes. Working: 100 × 50 × 8 = 40 000 bits. 40 000 ÷ 8 = 5000 bytes.

How the marks are given

- 1 mark: the working, 100 × 50 × 8 = 40 000 bits, then dividing by 8.
- 1 mark: the answer, 5000 bytes (5 KB also counts).
- The question asks you to show your working, so the answer on its own only gets 1 mark.

Watch out

- The question asks for bytes, so divide the bits by 8. 40 000 is the number of bits, not bytes.

2(g)

2 marks

Describe one difference between lossy and lossless compression.

The answer

Lossy compression permanently removes some data, so the file can't be put back exactly as it was. Lossless compression doesn't remove any data, so the original file can be rebuilt exactly.

How the marks are given

- 1 mark: you say lossy permanently removes data.
- 1 mark: you say lossless keeps all the data, so the original can be rebuilt.

Watch out

- Talk about both types, not just one.

3(a)

1 mark

State what is meant by a LAN.

The answer

A network that covers a small area, such as one building or one site.

How the marks are given

- 1 mark: a small area, one building or one site.

Watch out

- "A local network" doesn't get the mark. It just repeats the words.

3(b)

2 marks

A school uses a star topology. Explain one advantage of a star topology.

The answer

If one cable or device breaks, the rest of the network keeps working, because each device has its own cable to the central switch.

How the marks are given

- 1 mark: the advantage, for example the rest of the network keeps working if one cable fails.
- 1 mark: the reason, for example each device has its own connection to the switch.

Watch out

- For 2 marks, give the reason as well as the advantage.

3(c)

1 mark

Which protocol is used to send emails? Tick one box.

The answer

SMTP.

How the marks are given

- 1 mark: SMTP.

Watch out

- IMAP is for reading emails from the server, not sending them.

3(d)

2 marks

Describe what a DNS (Domain Name Server) does when a user types a web address into a browser.

The answer

The DNS server looks up the domain name and finds its IP address. It sends the IP address back to the browser, which uses it to ask the web server for the page.

How the marks are given

- 1 mark: you say it looks up the domain name and finds the IP address.
- 1 mark: you say it sends the IP address back, so the browser can request the page.

4(a)

2 marks

Describe what is meant by phishing.

The answer

Phishing is sending fake emails or messages that pretend to come from a trusted company, to trick people into giving away personal details like passwords, or into clicking a harmful link.

How the marks are given

- 1 mark: fake messages pretending to be from someone trusted.
- 1 mark: to trick people into giving personal details or clicking a link.

4(b)

2 marks

Explain how a firewall helps to protect a network.

The answer

A firewall checks the data coming into and going out of the network, and blocks anything that doesn't follow its rules, such as unauthorised access.

How the marks are given

- 1 mark: you say it checks the data coming in and going out.
- 1 mark: you say it blocks anything that doesn't meet its rules.

Watch out

- A firewall doesn't remove viruses. That's what anti-malware software does.

4(c)

2 marks

Identify one function of an operating system and describe what it does.

The answer

Memory management: the operating system shares out the RAM between the programs that are running, and moves data into virtual memory when RAM is full.

How the marks are given

- 1 mark: you name a function (memory management, multitasking, peripheral management, user management or file management).
- 1 mark: you describe what that same function does.

Watch out

- Your description must match the function you named.

5

2 marks

State two principles of the Data Protection Act 2018 that an online shop must follow with its customers' data.

The answer

Data must be kept secure. Data must not be kept for longer than it's needed.

How the marks are given

- 1 mark for each, up to 2 marks: used fairly, lawfully and openly; only used for the reason it was collected; only as much data as is needed; kept accurate and up to date; not kept longer than needed; kept secure; people can see the data held about them.

Watch out

- "You need permission" on its own isn't one of the principles.

Total: 30 marks. A rough guide for this practice paper: 22 or more is a strong result. 15 to 21 is solid, with some gaps to fix. Under 15 means going back over these topics in the notebook first. Whatever you got, write down the questions where you lost marks: those are what to revise next.